Shini Kim

ARP Assignment: Data Plan Proposal

1. Problem Overview and Research Question

Project Overview

The secondary market for vintage and boutique electric guitars (1950s–1980s) has evolved from a niche community into a high-stakes asset class. Unlike standard retail goods, these instruments do not follow linear depreciation, their value is dictated by a complex interplay of historical provenance, "player-grade" vs. "collector-grade" condition, and shifting cultural trends (ex. Kurt Cobain’s and Nirvana’s influence driving demand for Fender Jaguars). As a guitarist and AI student focused on retail demand forecasting, I aim to bridge the gap between subjective "vibe-based" pricing and data-driven inventory management.

Research Question

"To what extent can a multi-modal machine learning model, integrating time series sales data with unstructured text descriptions, accurately predict the 6-month price trajectory of vintage electric guitars in the North American secondary market?"

Importance of the Problem

For boutique retailers in hubs like Toronto, inventory overhead is the primary risk factor. Mispricing a vintage Stratocaster by even 10% can result in thousands of dollars of "dead stock" or lost margins. Accurately predicting which models are entering a "hype cycle" to these collectors and guitarists allows small business owners to optimize their capital allocation and stock instruments that will move quickly.

Justification for a Data-Driven Approach

The vintage guitar market is now largely digitized through platforms like Reverb and eBay, providing a rich repository of historical "Sold" listings. The volume of data (hundreds of thousands of listings) makes manual tracking impossible for a human, whereas a machine learning model can detect subtle correlations between macro-economic indicators, celebrity "signature guitar" releases, and used market fluctuations.

Alternative Approach (Non-Data Driven)

An alternative approach would be the Expert Appraisal Method, relying on a "Subject Matter Expert" (SME) to manually value instruments based on experience. I chose not to use this exclusively because human experts are very prone to cognitive biases, such as the endowment effect or geographic bias. A data-driven model provides an objective baseline that scales across thousands of SKUs simultaneously.

2. Data Source Identification

Data Types and Acquisition

I plan to utilize two primary data streams:

- **Reverb.com API / Price Guide Data:** This provides the ground truth for actual sold prices, listing dates, and item conditions (Mint, Excellent, Good, etc.).
- **Web-Scraped Descriptive Metadata**: Using tools like Beautiful Soup to extract unstructured text from vintage shop archives (ex. 12th Fret in Toronto, Emerald City Guitars) to capture nuanced features like "stock pickups included" or "refretted."

Reliability and Accessibility

Reverb's Price Guide is considered the industry standard for transaction transparency. While the API has certain rate limits for students, the historical "Sold" pages provide a consistent, structured format that is highly accessible for data extraction.

Choice of Source vs. Alternatives

I selected Reverb over eBay because Reverb is a specialized vertical market. eBay data contains too much noise (ex. guitar parts, toy guitars, or unrelated accessories) that would require excessive cleaning. Reverb's categorization is already tailored cleanly enough to musical instrument specifications.

Trade-off: Granularity vs. Sample Size.

I am choosing to focus only on "Electric Guitars" rather than all musical instruments. While this reduces the total volume of my dataset, it ensures higher quality and more relevant features (ex. pickup configurations: SSS, HH, HSH) which would be lost in a broader "Musical Equipment" model.

3. Data Description and Assessment

The data will be tabular in its final form, consisting of:

- **Numerical:** Year of manufacture, number of previous owners, price (USD/CAD).
- **Categorical:** Brand (Fender, Gibson), Model, Color/Finish, Condition Grade.
- **Textual:** Listing description (approx. 200–500 words per entry).

Potential Challenges

The most significant challenge is the subjectivity of condition. One seller's "Excellent" is another's "Very Good." This creates noise in the price to condition correlation.

Top Two Risks

1. **Feature Sparsity:** Vintage guitars are often modified (ex. changed tuners, replaced pickups). These modifications are often buried in the text rather than listed in a structured field, making it difficult for a model to quantify the purity of the instrument.
2. **Market Volatility/Outliers:** A single high-profile celebrity sale (ex. a Kurt Cobain-owned Jaguar) can create a spike in the data that does not reflect general market value.

If modifications aren't correctly identified, the model might predict a high price for a guitar that is actually devalued due to non-original parts, leading to poor inventory decisions for a retailer.

4. Data Preparation Plan

Steps and Reasoning

1. **Text Mining for "Originality":** Natural language processing will be used to scan descriptions for keywords like "non-original," "modded," or "OHSC" (Original Hard Shell Case). This turns unstructured text into a binary "Originality Score".
2. **Outlier Removal (Z-Score Filtering):** I will remove transactions that fall 3 standard deviations outside the mean for a specific year/model to eliminate "celebrity-owned" outliers that skew the baseline.
3. **Currency Standardization:** Converting all historical sales to a single currency (USD) at the historical exchange rate of the sale date to maintain temporal consistency.

Risks of Incorrect Preparation

If Step 1 (Text Mining) is done incorrectly: for example, if the model fails to recognize the word "not" in "not original", it will incorrectly flag a modified guitar as "Original," causing a massive overestimation in the predicted value.

Subjective Decision

Condition Normalization: I must decide how to weigh "Player Grade" instruments (vintage guitars with many repairs but high playability). I have chosen to treat them as a separate categorical cluster rather than just a lower-tier "Good" condition, as they often are a premium in the current "Grunge/Punk" market.

5. Ethical Considerations and Limitations

Ethical Risks:

- **Market Manipulation:** There is a risk that a highly accurate price prediction model could be used by large-scale resellers to sweep the market of undervalued listings, effectively pricing out individual musicians.
- **Bias in Valuation:** If the training data is sourced only from high-end US retailers, it may undervalue unique instruments from the Japanese or Korean markets (ex. vintage Fernandes or Greco guitars), which I personally value as a musician.

To address these, I will include a fairness weight in the model that ensures transparency in how "hype" vs. "utility" is calculated. I will also document the data sources to acknowledge the North American geographic bias.

6. Visualization and Communication Plan

Presentation Method

I will develop an Interactive Market Dashboard using Power BI, allowing users to filter by specific guitar models and see the predicted value versus the current market asking price. In order to enhance clarity, the dashboard will have dynamic time series line charts that showcase historical sold data with the AI-projected forecast, allowing users to toggle between different currency views and regional market filters.

Intended Audience

The primary audience consists of independent guitar shop owners in Toronto and serious collectors who need to justify their capital investments to stakeholders or partners. These users are often highly knowledgeable about instrument quality but may lack the quantitative tools required to present a rigorous, data-backed business case for high-value acquisitions.

Supporting Decision-Making

The output will include a "Buy/Hold/Sell" indicator based on the predicted 6-month trend. This supports inventory optimization by highlighting which items are likely to become stagnant stock.

Preventing Misinterpretation

- **Confidence Intervals:** I will display a shaded uncertainty range around every price prediction to indicate the model's margin of error. Without this visual guardrail, a user might interpret a single figure (ex. $4,500) as a guaranteed sale price rather than a statistical probability based on current market volatility.
- **Data Freshness Alerts:** I will also include a "Last Updated" timestamp on all visualizations. This ensures that users do not misinterpret a historical trend as current reality if the scraper has not accounted for a sudden, overnight shift in market demand.

7. References and Appendix

Reverb. (2026). *Reverb price guide: Real-time gear market data and historical transaction logs*. https://reverb.com/price-guide

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